Extra Bite #16: Antenna Patterns and Modeling

Reading Figures E9-1 and E9-2, far field vs near field, isotropic comparison, Method of Moments.

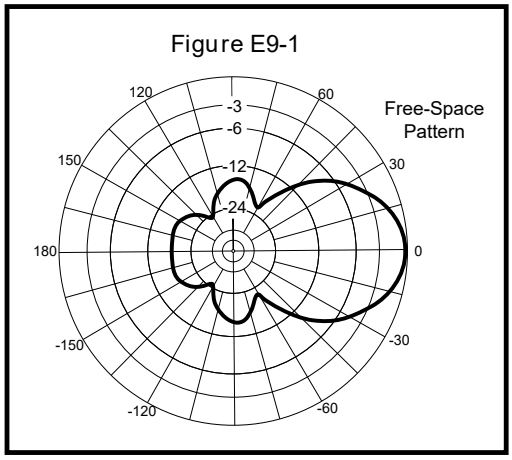
Group E9B · 11 pool questions · Standalone · 5-10 minute presentation · Practice exams at eham.net/exams

Reading Antenna Pattern Plots

Two of the official Extra figures are antenna patterns. E9-1 is an **azimuth** pattern (top-down view, beam directionality around the horizon). E9-2 is an **elevation** pattern (side view, how high the lobes ride above the horizon).

Three numbers come straight off these polar plots: **3 dB beamwidth** (how wide is the main lobe at the −3 dB rings), **front-to-back ratio** (main lobe vs the response 180° behind it), and **front-to-side ratio** (main vs the side response at 90°). Each ring on these plots is typically a known dB step (often 5 dB or 6 dB per ring), so reading values is just counting rings.

Figure E9-1 (Azimuth)



Three exam questions read directly from Figure E9-1. Memorize the values:

- **3 dB beamwidth = 50°.** (E9B01) Measured at the points on the main lobe where the response drops 3 dB below the peak.
- **Front-to-back ratio = 18 dB.** (E9B02) Main lobe minus the response directly behind (180°).
- **Front-to-side ratio = 14 dB.** (E9B03) Main lobe minus the response at 90° off-axis.

Figure E9-2 (Elevation)

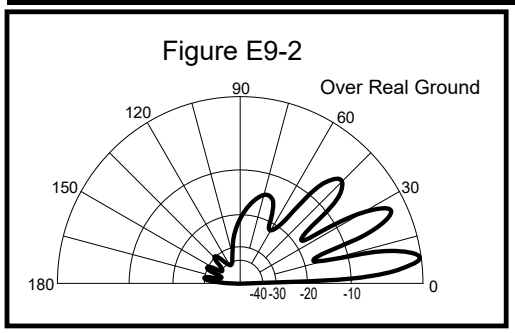


Figure E9-2 is an elevation pattern (E9B05). Three values from it:

- **Pattern type = elevation.** (E9B05) The horizontal axis is the horizon; the lobe rides up at some angle.
- **Peak elevation angle = 7.5°.** (E9B06) Low-angle DX-friendly takeoff.
- **Front-to-back ratio = 28 dB.** (E9B04) Main lobe versus the response behind the antenna.

Gain, Isotropic Reference, And Far Field

Question	Answer	Q
Difference in **radiated** power between a lossless antenna with gain and an isotropic radiator driven by the same power	**They are the same.** (E9B07) Gain redistributes power into a preferred direction; total radiated power is unchanged. (Higher gain in the main lobe is paid for by lower power elsewhere.)	E9B07
What is the **far field** of an antenna?	**The region where the shape of the radiation pattern no longer varies with distance.** (E9B08) Beyond a few wavelengths, the pattern locks in and only the field strength decreases with distance.	E9B08

Antenna Modeling: Method of Moments

Question	Answer	Q
Common modeling method	**Method of Moments** (E9B09). The basis of NEC and EZNEC.	E9B09
Principle of Method of Moments	**A wire is modeled as a series of segments, each having a uniform value of current** (E9B10). The numerical solver finds the segment currents that satisfy Maxwell's equations at the chosen frequency.	E9B10
Disadvantage of fewer than 10 segments per half-wavelength	**The computed feed point impedance may be incorrect** (E9B11). Under-segmenting compromises the accuracy of currents and therefore the calculated Z at the feed.	E9B11

Mental Cheat Sheet

- Figure E9-1 (azimuth): beamwidth **50°**, F/B **18 dB**, F/S **14 dB**.
- Figure E9-2 (elevation): peak elev angle **7.5°**, F/B **28 dB**.
- Lossless antenna with gain radiates the **same total power** as isotropic; gain just redistributes it.
- Far field = the region where the pattern shape stops changing with distance.
- Method of Moments is the standard antenna-modeling technique.
- MoM models a wire as segments, each with a **uniform current**.
- Use at least **10 segments per half-wavelength** for accurate Z.

Common Traps

- Antenna **gain** does NOT mean more total radiated power; it means the same power packed into a narrower lobe.
- Far field is about pattern **shape**, not field strength being constant.
- MoM segments hold **uniform current**, not “uniform voltage” or “a single sine generator.”
- Under-segmenting messes up **impedance**, not “harmonic radiation” or “mechanical stability.”
- Figure E9-1 is **azimuth**; E9-2 is **elevation**. The pool labels each pattern, but recognize both.

Takeaway: Antenna patterns are polar plots you read directly: beamwidth at −3 dB, F/B at 180°, F/S at 90°. Gain just rearranges a fixed total radiated power. Far field is where the pattern locks in. Method of Moments is how modern antenna modeling software works, slicing each wire into ten or more segments per half-wavelength so the computed feed impedance is trustworthy.

Bite 16 Question Bank (E9B)

E9B01. What is the 3 dB beamwidth of the antenna radiation pattern shown in Figure E9-1?
A. 75 degrees.
B. 50 degrees.
C. 25 degrees.
D. 30 degrees.

E9B02. What is the front-to-back ratio of the antenna radiation pattern shown in Figure E9-1?
A. 36 dB.
B. 14 dB.
C. 24 dB.
D. 18 dB.

E9B03. What is the front-to-side ratio of the antenna radiation pattern shown in Figure E9-1?
A. 12 dB.
B. 24 dB.
C. 18 dB.
D. 14 dB.

E9B04. What is the front-to-back ratio of the radiation pattern shown in Figure E9-2?
A. 15 dB.
B. 28 dB.
C. 3 dB.
D. 38 dB.

E9B05. What type of antenna pattern is shown in Figure E9-2?
A. Elevation.
B. Azimuth.
C. Near field.
D. Polarization.

E9B06. What is the elevation angle of peak response in the antenna radiation pattern shown in Figure E9-2?
A. 45 degrees.
B. 75 degrees.
C. 7.5 degrees.
D. 25 degrees.

E9B07. What is the difference in radiated power between a lossless antenna with gain and an isotropic radiator driven by the same power?
A. The power radiated from the directional antenna is increased by the gain of the antenna.
B. The power radiated from the directional antenna is stronger by its front-to-back ratio.
C. They are the same.
D. The power radiated from the isotropic radiator is 2.15 dB greater than that from the directional antenna.

E9B08. What is the far field of an antenna?
A. The region of the ionosphere where radiated power is not refracted.
B. The region where radiated power dissipates over a specified time period.
C. The region where radiated field strengths are constant.
D. The region where the shape of the radiation pattern no longer varies with distance.

E9B09. What type of analysis is commonly used for modeling antennas?
A. Graphical analysis.
B. Method of Moments.
C. Mutual impedance analysis.
D. Calculus differentiation with respect to physical properties.

E9B10. What is the principle of a Method of Moments analysis?
A. A wire is modeled as a series of segments, each having a uniform value of current.
B. A wire is modeled as a single sine-wave current generator.
C. A wire is modeled as a single sine-wave voltage source.
D. A wire is modeled as a series of segments, each having a distinct value of voltage across it.

E9B11. What is a disadvantage of decreasing the number of wire segments in an antenna model below 10 segments per half-wavelength?
A. Ground conductivity will not be accurately modeled.
B. The resulting design will favor radiation of harmonic energy.
C. The computed feed point impedance may be incorrect.
D. The antenna will become mechanically unstable.

Answer Key

E9B01: B	E9B02: D	E9B03: D	E9B04: B
E9B05: A	E9B06: C	E9B07: C	E9B08: D
E9B09: B	E9B10: A	E9B11: C	